



本周内容预览:

- 1. 真题
 - 2. 期末全部都是单选!
- 下面有部分选项是多选 (拼接了选项)

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Color Space

Which one of the following statements about colour spaces is incorrect?

- A. The R, G, and B channels of the RGB colour space are often correlated.
- B. The H and the S channel of the HSV colour space are confounded.
- C. The Y channel of the YCbCr colour space represents the brightness.
- D. The a* channel of the L*a*b* colour space is the green-blue component.

Which one of the following colour spaces is most perceptually uniform?

- RGB, HSV, YCbCr, Lab

Point Operations

Question: Given the following input image I (on the left) and output image O (on the right):

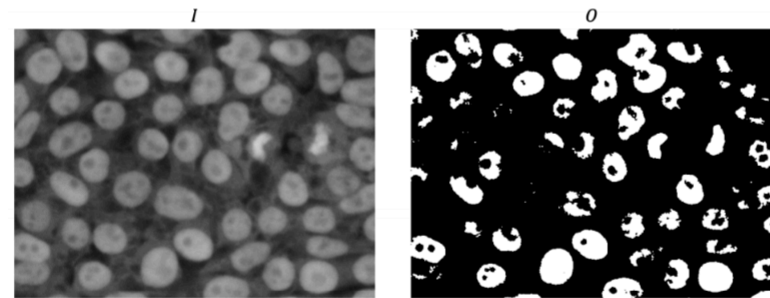
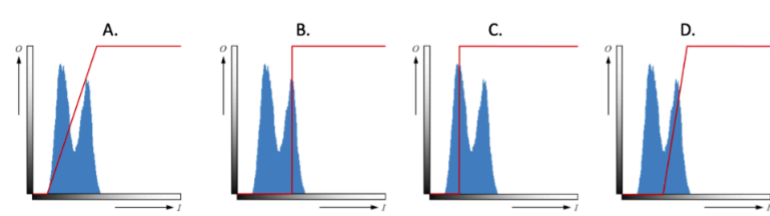


Image O is the result of processing image I using a point operation. Which one of the following mapping functions (red curves) from intensities in I to intensities in O was used by the point operator? For reference, in each graphic, the histogram of the image I is shown in blue.



哪个分割算法有以下的描述?

- 迭代计算初始阈值以上 (含) 和以下像素的平均强度, 并将这些平均值作为新阈值, 直到收敛。



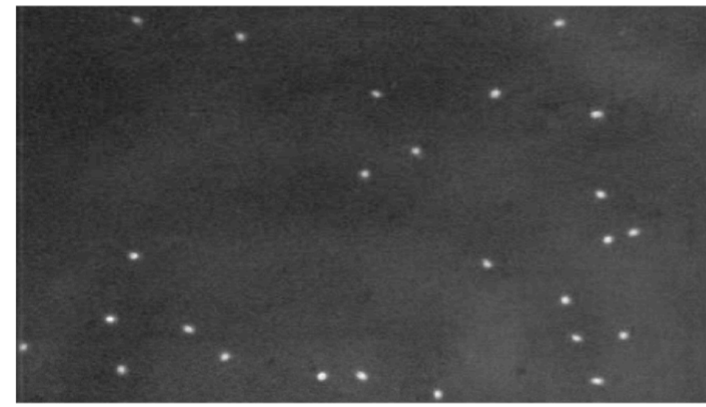
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Which two of the following statements about intensity transformations is incorrect?

- A. Contrast stretching linearly maps intensities between two values to the full output range.
- B. Log transformation maps a narrow range of high intensities to a wider range of output values.
- C. Power transformation can map intensities similar to log and inverse log transformations.
- D. Piecewise linear transformations can achieve contrast stretching and intensity slicing.
- E. Power transformation represents a family of transformations including log transformation.

Question (3 mark): Given the image below containing about two dozen algae (the bright spots) which we would like to segment using an automatic thresholding method. Which automatic thresholding method will give the best segmentation result here?



- Mean thresholding (计算整个图像的平均亮度, 并将其作为阈值)
- IsoData thresholding.
- Otsu thresholding.
- Triangle thresholding.

图像特征: 图中亮点 (藻类) 与暗背景对比鲜明。这意味着直方图根可能在较低的灰度值 (背景) 集中有一个峰值, 而较高的灰度值 (藻类) 则相对较少但明显分布。

Histogram Equalization

- Enhances contrast for intensity values near histogram maxima and decreases contrast near histogram minima
- CDF: cumulative distribution function (cumulative histogram)
- A single mapping function



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Neighborhood Operations

Question: Which one of the following equations concerning convolution (*) of two images (f and g) is correct?

- A. $f * g = (g * f)^{-1}$
- B. $f * (g + h) = (f * g) + (f * h)$
- C. $a \cdot (f * g) = (a \cdot f) * (a \cdot g)$
- D. $f * g = \hat{f} * \hat{g}$

Which one of the following image processing operations does not result in noise reduction?

- A. Applying median filtering with a kernel of radius 3 pixels.
- B. Applying averaging of multiple images of the same scene.
- C. Applying local averaging with a uniform kernel.
- D. Applying histogram equalization to the image.

Question: Suppose we have a 2D image that is very noisy and we want to smooth it by filtering with a uniform kernel of size 5×5 voxels. Let S be the total number of floating-point operations required per pixel to perform the operation when using a dimensionally separated implementation of the filter. And let N be the total number of floating-point operations required per pixel to perform the operation when using a dimensionally nonseparated implementation of the filter.

What is the ratio S/N (as a percentage rounded to one decimal)?

- A. 36.7%
- B. 40.0%
- C. 58.8%
- D. 66.7%

如果 Kernel size = $7 \times 7 \times 7$, 上面的比率会变成多少?

在运行水平方向二阶导数 [1,-2,1] 两次, 与其效果相同的四阶导数是什么?

Suppose you are given a 2D image I and you are asked to implement an algorithm that computes the edge image | ∇I |. On some website, you find the following four-step algorithm that supposedly accomplishes that operation. Which step is incorrect?

- A. Convolve I with a first-order derivative kernel in x and in y.
- B. Square the resulting images from the previous step pixelwise.
- C. Multiply the resulting images from the previous step pixelwise.
- D. Square-root the resulting image from the previous step pixelwise.



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Fourier transform

Which one of the following statements about frequency-domain processing is incorrect?

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- B. Noise suppression in images can be obtained by removing high-frequency components.

Fourier transform

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- B. Noise suppression in images can be obtained by removing high-frequency components.
- C. Convolution in the frequency domain corresponds to **correlation** in the spatial domain.
- D. Changing one pixel in the frequency domain may affect all pixels in the spatial domain.

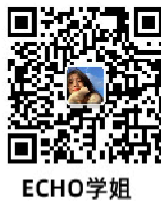
A. Any digital image can be written as the sum of a finite number of frequencies.
B. Noise suppression in images can be obtained by removing high-frequency components.

- Which one of the following statements on image filtering is incorrect?

B. Low-pass filtering results in blurry images.

C. High-pass filtering smooths fine details in images.

D. Notch filtering removes specific image frequencies.



Features

为什么直接使用 pixel?

- 像素值随光线强度、颜色和方向变化，并且有太多冗余特征的意义
- 可重复的 Reproducibility (robustness): 特征应该在不同的图像中同样位置上可检测，即使光照和视角发生变化，也能检测到相同的特征。
- 显著的 Saliency (descriptiveness): 每个特征应该有独特且可匹配的描述，使其在图中易于识别和匹配。
- 紧凑性 Compactness (efficiency): 特征应该是少量的和紧凑的，以提高计算效率和减少存储需求。

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Corresponding example co-occurrence matrices:

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$P_{(1,0^\circ)} =$

18	6	1	0
6	14	8	0
1	8	19	10
0	0	10	14

$P_{(1,90^\circ)} =$

18	6	2	
6	10	8	
2	8	14	
0	2	11	

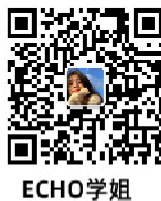
SIFT calculates feature vectors with lengths depending on image complexity?

Which **two** of the following statements about feature descriptors is incorrect?

- A. Haralick features are derived from gray-level co-occurrence matrices.
- B. SIFT achieves rotation invariance by computing gradient histograms at multiple scales.
- C. LBP describes local image texture and can be multiresolution and rotation-invariant.
- D. Colour moments have lower representation capability than the colour histogram.
- E. SIFT achieves scale invariance by computing gradient histograms at multiple scales.

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BoW

将所有的局部图像特征转化成固定维度的直方图。

将可变的训练图像，提取局部 SIFT keypoint descriptors。

通过 K-means 聚类算法，将得到的局部特征描述子聚类聚集（K）的簇，每个簇的中心点代表一个视觉单词（visual word）。

将每个特征描述子分配给距离最近的 visual word。

简单几何形状描述符

$Circularity = 4 * \pi * \text{面积} / (\text{周长的平方})$

对于所有的训练图像, 提取局部 SIFT keypoint descriptors。

通过 K-means 聚类算法, 将得到的局部特征描述子分组聚类, 以创建固定数量 (K) 的簇。每个簇的中心点代表一个视觉单词 (visual word)。

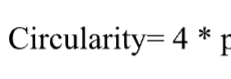
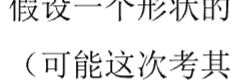
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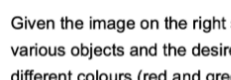
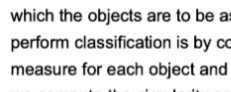
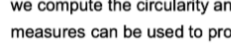
简单几何形状描述付

$Circularity = 4 \cdot \pi \cdot \frac{\text{面积}}{\text{周长}^2}$ (周长的平方)

假设一个形状的面积和周长都是 25，求它的 Circularity。

(可能这次考虑其它描述度的计算)

Given the image on the right showing the result of a segmentation of various objects and the shown classification of these objects. The two different colors (red and green) indicate the two different classes which the objects need to be assigned to. In order to be able to perform classification it is necessary to compare the objects. To perform classification is by comparing the value of a quantitative shape measure for each object and then thresholding these values. Suppose we compare the circularity and the compactness. Which of these two measures can be used to process the shown classification?

假设一个形状的面积和周长都是 25, 求它的 Circularity.

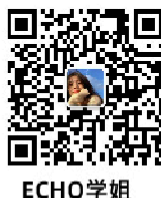
(可能这次考其它描述符的计算)



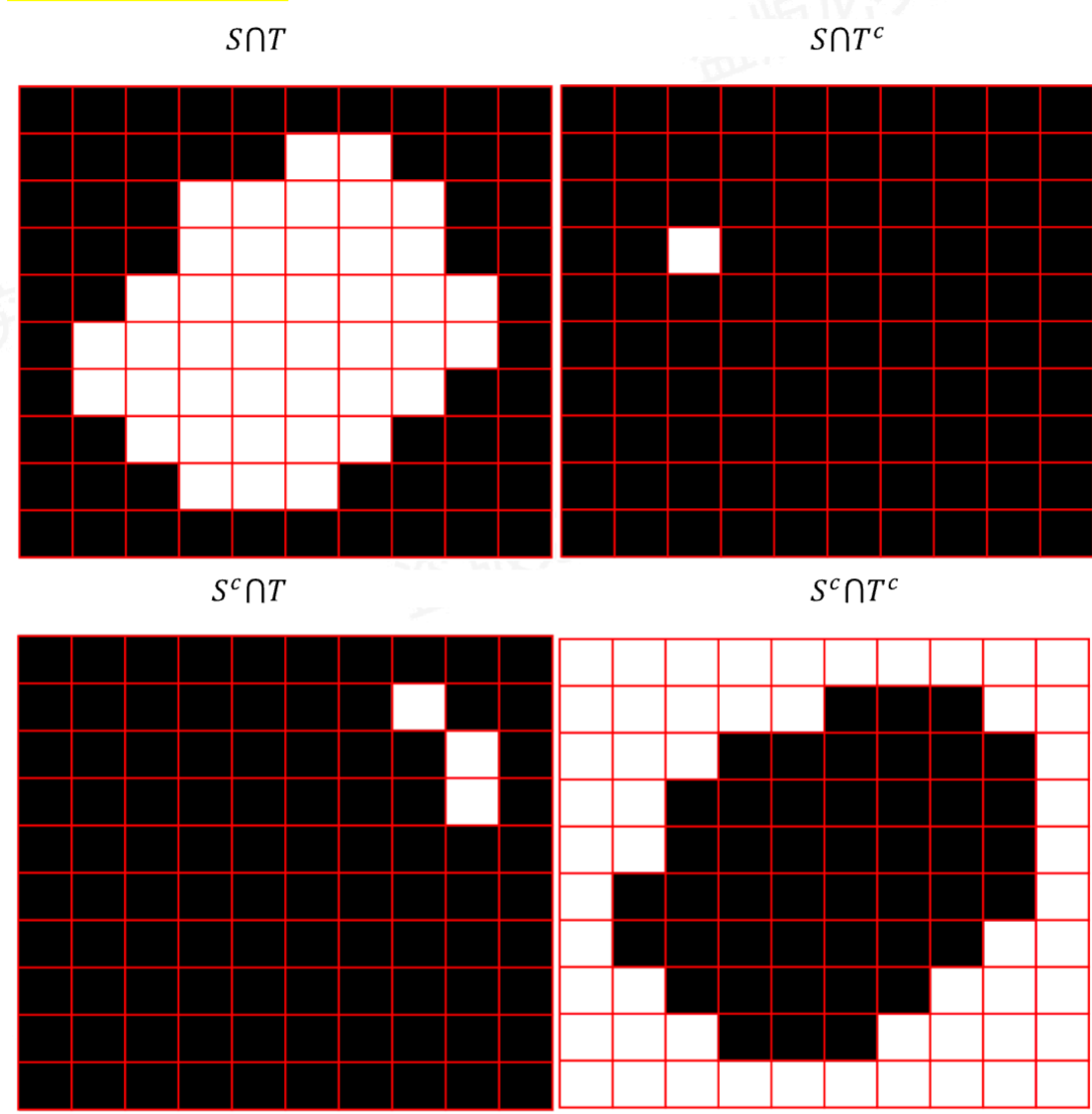
- A. Only circularity
B. Only eccentricity
C. Both circularity and eccentricity
D. Neither circularity nor eccentricity

A. Shape features are designed to be invariant to changes in viewpoint and illumination.

- B. Simple shape descriptors such as circularity are typically extracted after image segmentation.
- C. Shape context is computed from the histogram of gradients in a log-polar coordinate system.
- D. HOG descriptor requires segmentation to find the window for calculating gradient orientations.

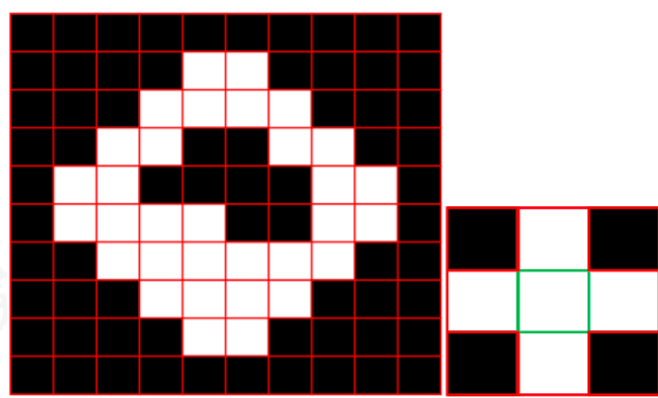


有二值图像 S 和 T ，一个是预测结果，一个是真实结果。下面是一些运算结果，JSC 是多少？

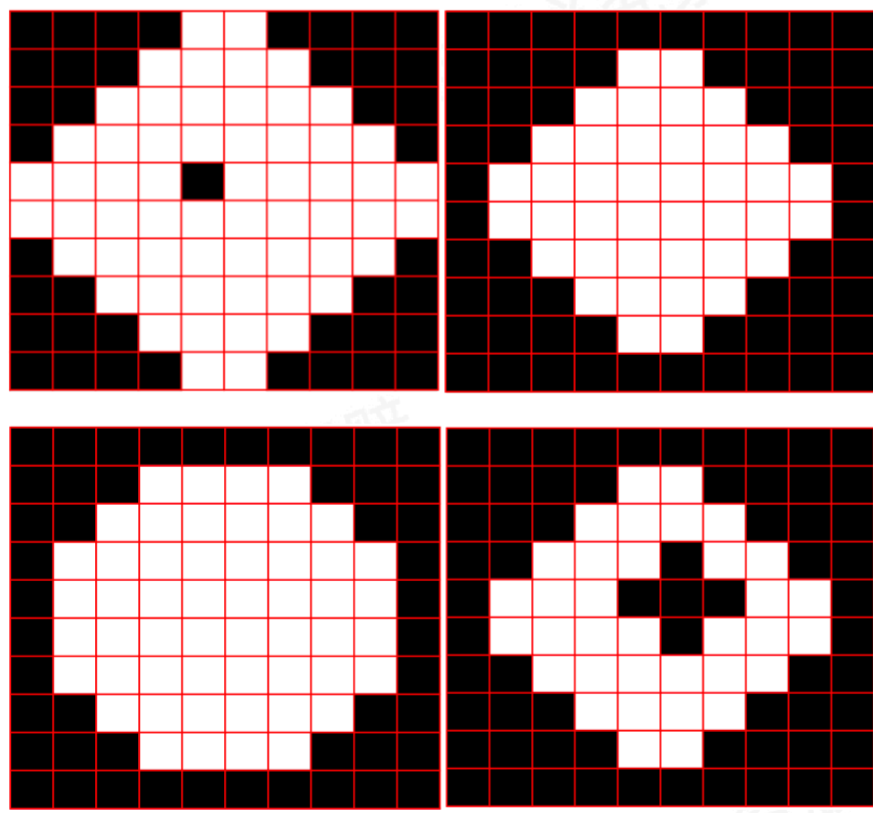


形态学

给一个二值图，还有结构元素



中心像素是原点，请问下面哪个是做膨胀之后的结果！

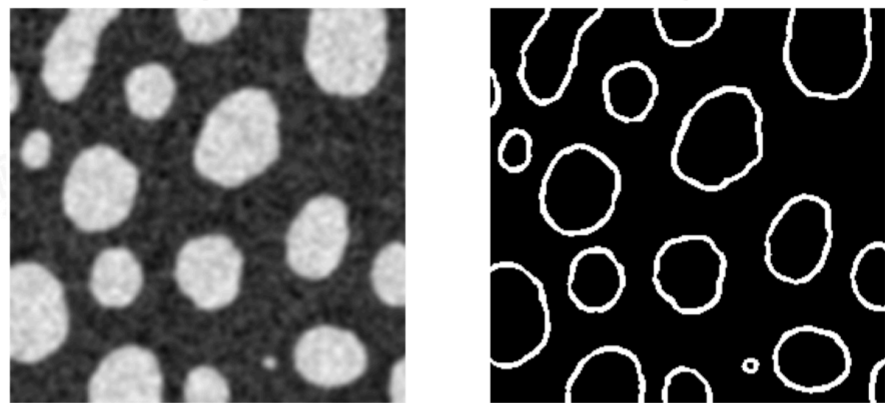


形态学梯度

- 基础梯度：基础梯度是用膨胀后的图像减去腐蚀后的图像得到差值图像。
- 内部梯度：是用原图像减去腐蚀之后的图像得到差值图像，称为图像的内部梯度。
- 外部梯度：图像膨胀之后再减去原来的图像得到的差值图像，称为图像的外部梯度。



Question: Suppose we have the following grayscale input image I (shown on the left) and we would like to have a binary output image O (as shown on the right) containing the edges of the objects in the input. Also suppose that the only image processing operations available for this are intensity thresholding, dilation and erosion (binary and grayscale) and image subtraction. Which one of the following algorithms will not produce the desired output?

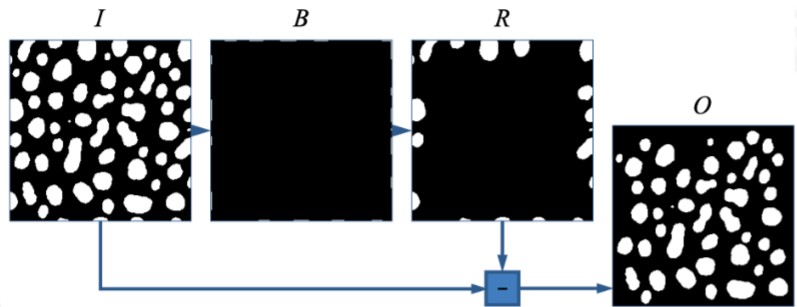


- A. Thresholding (result = T), then erosion (result = E), then subtraction (result = $E - T$).
B. Erosion (result = E), then subtraction (result = $I - E$), then thresholding (result = T).
C. Thresholding (result = T), then dilation (result = D), then subtraction (result = $D - T$).
D. Dilation (result = D), then subtraction (result = $D - I$), then thresholding (result = T).



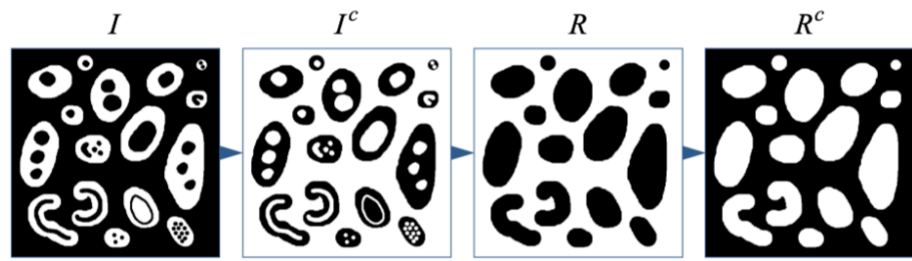
移除部分在图像中的对象

Take the boundary pixels B of input image I as the seeds, compute the reconstruction R from those seeds, and subtract the result from the input



填充图像中所有对象的孔洞

Take the complement I^c of image I , take the boundary pixels of I^c as seeds, compute reconstruction R of I^c from those seeds, take the complement R^c



Boundary: 从上下左右四个边界开始重建！

Given a binary input image I to which we apply the following algorithm:

- Step 1: Create a copy C of input image I .
Step 2: Copy the boundary pixels of C into new image B .
Step 3: Compute the reconstruction R of C from B .
Step 4: Compute output O by subtracting R from I .

What does the output image O contain?

- A. The same objects as the input image whatever the input.
B. The same objects as the input image but with holes filled.
C. The same objects as the input image except the boundary objects.
D. The same objects as the input image but with touching objects separated.

灰度膨胀可以通过使用一个灰度结构元素对灰度图像进行局部最大值滤波来实现。（去除小黑点）

灰度腐蚀可以使用一个灰度结构元素对灰度图像进行局部最小值滤波来实现。（去除小亮点）



CNN

较大的步幅 stride 会加速计算，但会减少输出特征图的尺寸。

Loss function

- A loss function is used to compute the model's prediction accuracy from the output
- The training objective is to minimise this loss
- The loss guides the backpropagation process to train the CNN model
- Gradient descent based methods, such as Stochastic gradient descent and the Adam optimizer, are commonly used algorithms for optimisation

验证集用于监控模型的性能，并进行超参数调优。主要目的是防止模型过拟合，在训练过程中监控模型的泛化能力。测试集和防止过拟合没有关系。

Which one of the following statements about data augmentation for image classification using CNNs is correct?

- A. Augmentation increases the number of parameters in the model.
B. Augmentation reduces the size of the dataset to speed up training.
C. Augmentation generates new training data by transforming the existing data.
D. Augmentation reduces the depth of the network for better performance.

Which one of the following is **not** the purpose of convolutional layers in CNNs?

- A. Reducing the number of parameters.
B. Convolving images with kernels.
C. Applying nonlinear activation function.
D. Making the network learn image features.

What is the purpose of convolutional layers in CNNs?

- A. Reducing the size of the feature maps. (pooling)
B. Calculating the dot product of the input and kernels.
C. Applying a nonlinear activation function. (after convolutional layer)
D. Making the network learn faster. (algorithm, architecture)

